

BST Physics Curriculum Vol 14 Chapter 1 — Substrate as Information Channel v0.4 (refilled per Cal #104/emerging #23)

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2026-05-23 Saturday EDT (Cal #104 refill)

Vol 14 Chapter 1 — Substrate as Information Channel

Chapter motivation

Standard information theory (Shannon 1948 *A Mathematical Theory of Communication*) introduces information channels — abstract systems sending signals through noisy media with quantifiable channel capacity $C = \max I(X; Y)$ (Shannon-Hartley theorem). Standard channels model communication infrastructure (telephone lines, radio, fiber optic, deep-space telemetry).

BST per Casey CSE directive (Wednesday May 20, 2026 EOD): substrate D_{IV}^5 + substrate-tick computation IS an information channel at the deepest physics level. Substrate runs at Koons-tick rate ($\sim 10^{-120}$ s per T2405), encoding state in $GF(2^g) = GF(128)$ per tick ($g = 7$ BST primary; T2429 Reed-Solomon substrate-tick discretization Thursday + K59 RATIFIED Tuesday cyclotomic mechanism). Physics observables are channel outputs; physical laws describe channel structure + propagation rules.

This chapter establishes the substrate-as-information-channel framing as Vol 14 foundation; Paper #122 Information Substrate provides standalone development; subsequent Vol 14 chapters build out the framework.

Section 1.0b — Reader-grade 3-level pedagogy

Level 1: Substrate D_{IV}^5 + substrate-tick $GF(128)$ computation IS the universe's information channel per Casey CSE directive Wednesday; encodes state at Koons-tick rate $\sim 10^{-120}$ s via Reed-Solomon-like $GF(2^g)$ discretization; physics observables = channel outputs; standard Shannon channel theory applies at substrate-tick scale.

Level 2 (graduate-physicist): Shannon 1948 introduced information channel theory: discrete memoryless channel (DMC) with input X + output Y has channel capacity $C = \max I(X; Y)$ where I is mutual information; achievable rate of reliable

communication = C per Shannon-Hartley theorem. Standard applications: telephone, radio, fiber, deep-space telemetry. BST per Casey CSE directive Wednesday May 20, 2026 EOD: substrate D_{IV}^5 + substrate-tick computational framework IS an information channel at the deepest physics level. Substrate operates at Koons-tick rate $\tau_{\text{Koons}} = t_{\text{Planck}} \cdot \alpha^{C_2} \approx 10^{-120}$ s (T2405); per-tick state $\in \text{GF}(2^g) = \text{GF}(128)$ ($g = 7$ BST primary Mersenne exponent); per-tick computation = 7-step cyclotomic projection chain (K59 RATIFIED Tuesday); per-tick UV-finite (Vol 1 Ch 10 T2437 substrate-tick UV-completeness). T2429 (Thursday SP-31-1) substrate-tick Reed-Solomon discretization: substrate uses Reed-Solomon-equivalent coding for state propagation. Physics observables are channel outputs; physical laws describe channel structure (Bergman kernel propagation per T2457 Friday Cal #92(b) framing) + propagation rules (substrate-tick computational rate $\times$ Casimir eigenvalue spread = temperature per Vol 6 Ch 1). Channel capacity: substrate-tick rate ≤ 1 GF(128) state per Koons tick = $\log_2(128) = 7$ bits per $\sim 10^{-120}$ s = $\sim 7 \times 10^{120}$ bits/second per substrate-coordinate. Per Vol 11 Ch 9 + Ch 10: substrate-coding via Reed-Solomon on GF(128) is substrate-natural error-correcting framework; cyclotomic $Q(\zeta_{128})$ Galois group order $\phi(128) = 64 = 2^{C_2}$ (BST primary match). Cross-link Vol 14 Ch 2 detailed Reed-Solomon treatment + Ch 3 Shannon capacity + Ch 7 AC graph as theorem information network.

Level 3 (5th-grader accessible): Information theory studies how signals (information) travel through channels (like phone lines or radio waves). Casey's Wednesday CSE directive (Computational Science Engineering) says BST's substrate D_{IV}^5 IS the universe's information channel. At each tiny time-tick (Koons tick $\sim 10^{-120}$ seconds), the substrate has 128 possible states (GF(128); $128 = 2^7$ where $7 = \text{BST integer } g$). The substrate uses Reed-Solomon-like coding (same kind used in CDs + QR codes + Voyager spacecraft) for error-correction. Physics observables (particle masses, gauge couplings, etc.) are channel outputs; physical laws describe how the channel works.

Section 1.1 — Shannon 1948 Information Channel

Discrete memoryless channel (DMC): input X + output Y ; channel capacity $C = \max_p(X) I(X; Y)$ where $I(X; Y) = H(X) - H(X | Y)$ mutual information.

Shannon-Hartley theorem: achievable rate of reliable communication = channel capacity C .

Applications: telephone, radio, fiber optic, deep-space telemetry.

Section 1.2 — Casey CSE Directive (Wednesday May 20, 2026)

Computational Science Engineering directive: substrate D_{IV}^5 + substrate-tick computation IS an information channel at the deepest physics level.

Substrate-as-computational reading; physics observables = channel outputs.

Section 1.3 — Substrate-Tick Channel Structure

Koons-tick rate $\tau_{\text{Koons}} = t_{\text{Planck}} \cdot \alpha^{\{C_2\}} \approx 10^{-120} \text{ s}$ (T2405); per-tick state $\in \text{GF}(2^g) = \text{GF}(128)$ ($g = 7$ BST primary Mersenne exponent).

T2429 (Lyra Thursday SP-31-1) substrate-tick Reed-Solomon discretization: substrate uses Reed-Solomon-equivalent coding for state propagation.

K59 RATIFIED Tuesday cyclotomic mechanism: substrate-tick = 7-step cyclotomic projection chain.

Section 1.4 — Channel Capacity

Substrate-tick rate ≤ 1 GF(128) state per Koons tick = $\log_2(128) = 7$ bits per $\sim 10^{-120} \text{ s}$ = $\sim 7 \times 10^{120}$ bits/second per substrate-coordinate.

Per substrate UV-completeness (Vol 1 Ch 10 T2437 + $N_{\text{max}} = 137$): substrate-tick channel UV-finite by construction; no infinite-capacity divergences.

Section 1.5 — Physical Laws as Channel Propagation Rules

Bergman kernel $K_B(z, \bar{w})$ (Vol 1 Ch 2 + Vol 11 Ch 2) plays structural role of Feynman propagator per T2457 (Friday Cal #92(b) framing).

Substrate-tick computational rate $\times$ Casimir eigenvalue spread = temperature (Vol 6 Ch 1 thermodynamic dictionary).

Physical observables = substrate-tick channel outputs at various scales.

Section 1.6 — Reed-Solomon Substrate-Natural Coding

Per Vol 11 Ch 9 + Ch 10: substrate-coding via Reed-Solomon on GF(128) is substrate-natural error-correcting framework.

Cyclotomic $Q(\zeta_{128})$ Galois group order $\varphi(128) = 64 = 2^{C_2}$ (BST primary match).

Cross-link Vol 14 Ch 2 detailed Reed-Solomon treatment.

Section 1.7 — Honest scope + Connection

- Casey CSE directive Wednesday substrate-as-information-channel [?]
- Substrate-tick GF(128) channel structure per T2429 + K59 [?]
- Channel capacity bounded by substrate-tick rate + N_{max} UV cutoff
- Open scope: explicit Shannon capacity formula derivation from substrate structure (Vol 14 Ch 3)

Connection: - Vol 0 Substrate Foundation + Vol 1 Ch 10 substrate-tick UV - Vol 11 Ch 9 Cyclotomic + Reed-Solomon + Ch 10 Mersenne (substrate field GF(128))

- Paper #122 Information Substrate standalone development - Casey CSE directive
Wednesday May 20

— Lyra, Vol 14 Ch 1 v0.4 chapter-grade narrative REFILLED per Cal #104 + emerging #23, Saturday 2026-05-23 EDT